Google Developer Group
Bandung Institute of Technology

Team name	Tide Ledger
Project title	RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports
Selected case	Case 1: Smarter Campus and Learning
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Abstract

1. Problem. Untrained student treasurers manage campus funds under strict bylaws: physical stamps, balanced ledgers, meal-attendance ratios, revenue stamps over Rp5,000,000 (Law 10/2020), taxes, and deadlines. Incomplete submissions freeze campus funding and peer learning [6], trapping study hours in mechanical revisions without fiscal judgment.

2. Intended Users. Primary: club and BEM treasurers and secretaries (maintaining attendance and verifying figures). Secondary: faculty verifiers (Biro Kemahasiswaan, student auditors) conducting desk reviews to log audit trails without releasing funds.

3. Supporting Observations or Evidence. Siada and Tungadi (2021) document unstandardized activity documentation at BKAM UAJM [1]. Guidelines diverge: UB requires physical stamps and balanced ledgers [2]; UGM enforces Rp10,000 stamps above Rp5,000,000 under Law 10/2020 [3]; ITS mandates 10-day deadlines [4] and taxes [5]; FILKOM UB notes recurring withholding failures [6].

4. Proposed Solution and Core User Flow. RapiLPJ turns rejection into a gated practice run. (1) Perceive, then confirm. Gemini Flash extracts receipt metadata into structured JSON without arithmetic. Fields display beside images for treasurer confirmation; manual adjustments are preserved as an edit disclosure log for verifier review. (2) Deterministic kernel in portable .lpj: duplicate invoices, dates outside event windows, missing stamps over Rp5,000,000, and variances above 10%. Meal portions and PPh 21 evaluate against a rule card signed by two UAJM advisors whose reference hash is published out-of-band by faculty, preventing unilateral parameter tampering. Missing card rates return 'preset gap'; the engine never invents rates. (3) Gated literacy. Infractions lock export until micro-quizzes match statutory keys (Rp5,000,000 exactly is not dutiable; nominals above are dutiable under Law 10/2020, Art. 3). Templates draft variance addenda without LLM involvement. (4) Desk-check file, not disbursement. Export yields indexed PDF dossiers plus SHA-256 hashes binding confirmed ledgers, quizzes, and addenda. Verifiers recompute hashes locally against the out-of-band rule hash, review disclosed manual edits, and record desk-check clearance into .lpj files without releasing funds.

5. How It Differs from Existing Alternatives. Pure LLMs miscalculate ledgers; spreadsheets lack hash-binding, statutory checks, and quiz gates; portals merely collect files. RapiLPJ enforces a strict boundary: Gemini perceives, humans confirm figures, while arithmetic, statutory checks, quiz keys, and addenda remain 100% deterministic client-side code.

6. Planned Technologies. Perception: Gemini Flash on Vertex AI (Singapore, asia-southeast1) under institutional non-training terms, paired with offline Indonesian receipt fixtures for zero-setup evaluation. Kernel: open-source JavaScript, locked templates, FileReader, .lpj JSON, and SHA-256. Presentation: WCAG 2.1 AA accessible bilingual HTML5/CSS3 deployed statically on Firebase Hosting with zero application databases.

7. Expected Benefit and Development Plan. We evaluate 30 treasurers and 2 UAJM advisors across five metrics: treasurer minutes, mechanical defects before/after kernel, fields corrected after Gemini, un-gated retention quiz scores, and verifier minutes. Semifinal benchmarks single-receipt perception, confirmation, signed card, quiz gate, and desk-checks; Final Day delivers multi-page batching and QR re-checks.

Source Acknowledgments. [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119 (studi sistem informasi portofolio kemahasiswaan BKAM UAJM). [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026. AI disclosure: Claude and Gemini supported research, deterministic engine code, and drafting.